

Name : Student ID :

1. What is the difference between trap and interrupt? What is the use of each function?
2. What is the reason for provision of dual-mode of operations? State the side-effect of using it?
3. The following is a list of instructions that are normally protected. What is the minimal set of instructions that must be protected (state reason)?
 - a. Change to user mode.
 - b. Change to monitor mode.
 - c. Set the value of timer.
 - d. Turn on timer interrupt.
 - e. Turn off timer interrupt.
4. State three events on which a process leave the CPU and state the expected action taken by the operating system in each case?

5. Give advantages/disadvantages of selecting process quantum to be (Short or long).

6. Suppose that the following processes arrive for execution at the times indicated in the table. Each process will run the listed amount of time.

Calculate the average waiting, response and turnaround times in the following cases:
[3 Marks]

- Preemptive SJF.
- Non-preemptive SJF.
- RR.

process	Arrival time	Burst time
P ₁	0.0	10
P ₂	0.4	4
P ₃	1.0	1

	response	waiting	turnaround
P ₁			
P ₂			
P ₃			
average			

Preemptive SJF

	response	waiting	turnaround
P ₁			
P ₂			
P ₃			
average			

Non-Preemptive SJF

	response	waiting	turnaround
P ₁			
P ₂			
P ₃			
average			

RR